



**Northern Beaches Secondary College
Manly Selective Campus**

**2011
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION**

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators and templates may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1- 10
- All questions are of equal value

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Question 1 (Answer in a separate booklet)

12

- (a) Expand and simplify $2(4x - 1) - 13x$ (1)
- (b) If the numbers 7, 1 and -5 are the first 3 terms of an arithmetic series, show the n th term is given by $T_n = 13 - 6n$ (2)
- (c) Solve $|2x - 5| > 3$ and graph the solution on a number line. (3)
- (d) Differentiate $y = \cos^3 x$. (2)
- (e) Draw a graph of the function $y = -\sqrt{4 - x^2}$ showing x and y intercepts. (1)
- (f) Determine the value of a and b if $\frac{5}{2 + \sqrt{3}} = a + b\sqrt{3}$ (2)
- (g) Simplify $\log_5 125$ (1)

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Question 2 (Answer in a separate booklet)

12

(a) Show that $x^2 - 4x + 2k = 0$ has real roots for $k \leq 2$. (2)

(b) Find an indefinite integral of

(i) $\int (\sqrt[3]{x} + x^2) dx$ (2)

(ii) $\int (e^{2x} + \sin 3x) dx$ (2)

(c) Find the co-ordinates of the vertex and the equation of the directrix for the parabola $y^2 = 8 - 4x$. (2)

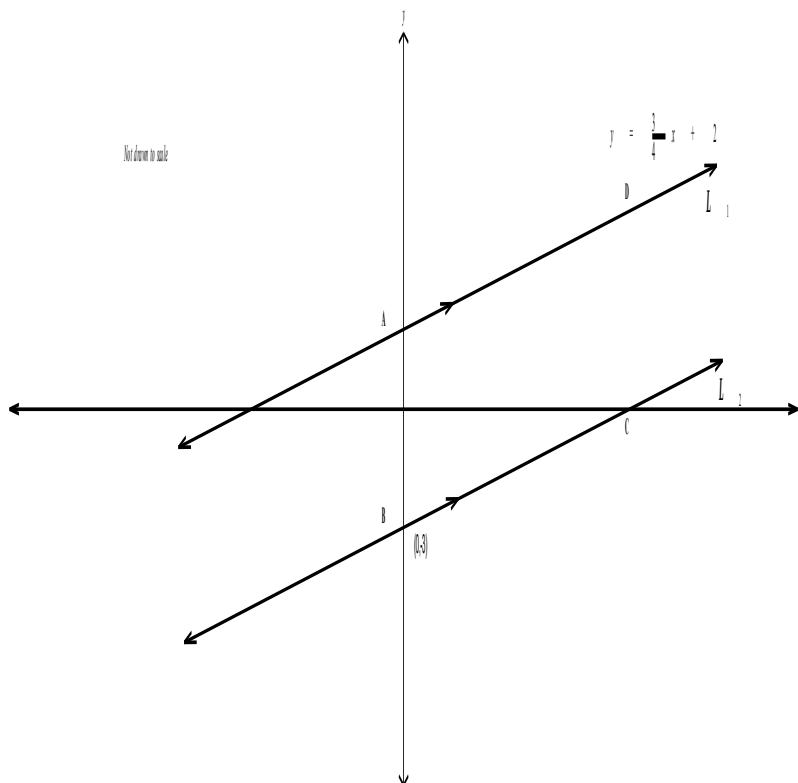
(d) The gradient of the tangent function of a curve is given by $\frac{dy}{dx} = 3 \sin 2x$.
If the curve passes through (0,0), find its equation. (2)

(e) Solve $\tan \theta = -2$ for $0 \leq \theta \leq 2\pi$. (2)

Question 3 (Answer in a separate booklet)

12

(a)



The line L_1 has equation $y = \frac{3}{4}x + 2$ and passes through the point A on the y -axis.

The line L_2 is parallel to L_1 and passes through $B(0, -3)$.

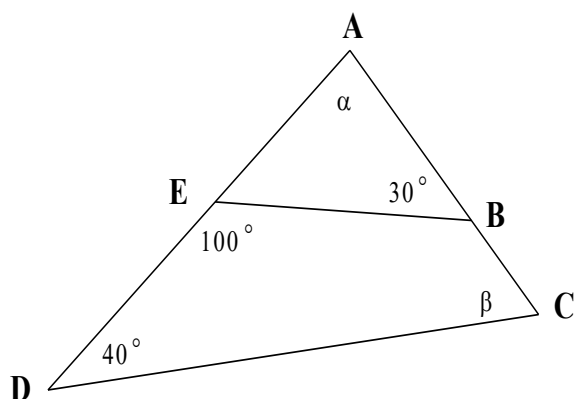
- (i) Write down the co-ordinates of A . (1)
- (ii) Find the equation of the line L_2 . (1)
- (iii) If C is the x -intercept of L_2 , write down the co-ordinates of C . (1)
- (iv) The point D is positioned on line L_1 , such that $ABCD$ is a rhombus.
Find the co-ordinates of D . (1)
- (v) Calculate the perpendicular distance from A to L_2 . (1)
- (vi) Hence, or otherwise, calculate the area of the rhombus $ABCD$. (2)

Question 3 continues on the next page.

Question 3 (continued)

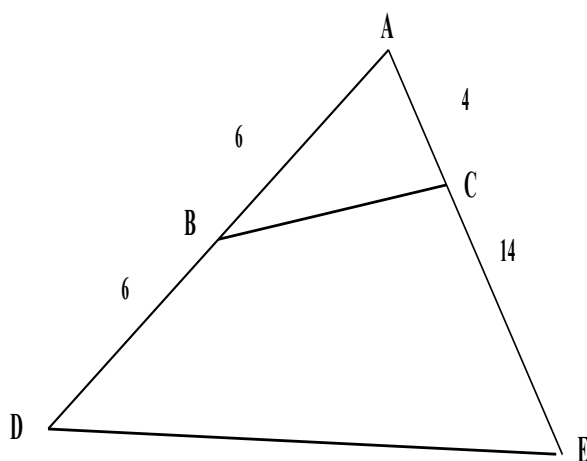
(b) Calculate the size of angles α and β giving reasons for your answer.

(2)



(c) Prove that $\triangle BAC \parallel \triangle DAE$

(3)



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Question 4 (Answer in a separate booklet)

12

- a) Determine the value of A, B and C in the following. (3)

$$2x^2 + 6x + 9 \equiv A(x + 2)(x - 1) + B(x + 2) + C$$

- b) The first three terms of a series are 3, 6 and 12 (*leave your answers in index form*).

(i) What is the value of T_8 ? (1)

(ii) What is the sum of the first fifteen terms? (1)

- c) The first multiple of seven after 130 is 133. What is the sum of all the multiples of seven between 130 and 500? (4)

- d) What is the maximum value for S_n ? (3)

$$S_n = 1 + \frac{\sin \theta}{2} + \frac{\sin^2 \theta}{4} + \frac{\sin^3 \theta}{8} + \dots$$

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Question 5 (Answer in a separate booklet)

12

- (a) The quadratic equation $2x^2 - 3x + 6 = 0$ has roots α and β .
Find the value of:

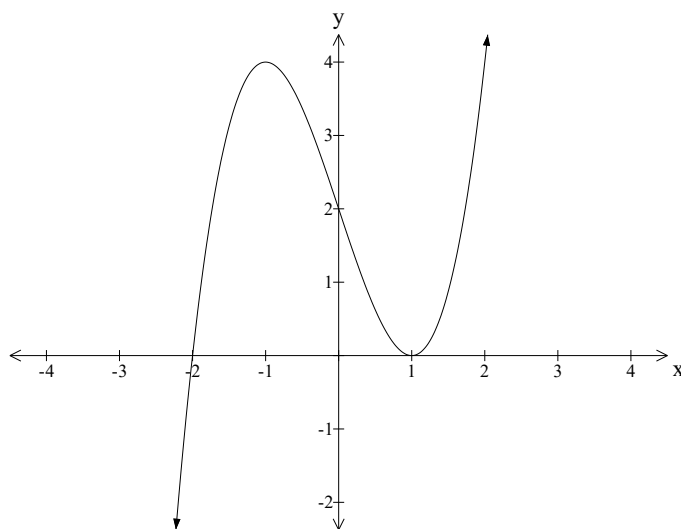
(i) $\alpha + \beta$ (1)

(ii) $\alpha\beta$ (1)

(iii) $\alpha^2 + \beta^2$ (2)

- (b) Differentiate $y = \sin 2x^\circ$. (2)

- (c) Copy the graph shown below into your answer booklet and, on the same set of axes, sketch the curve of the derivative function. (2)



- (d) Consider the curve $y = 12x^3 - 3x^4$.

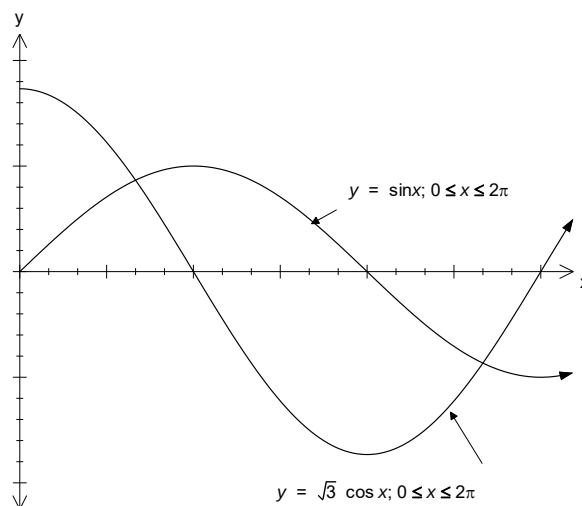
Find the stationary points and determine their nature. (4)

Question 6 (Answer in a separate booklet)

12

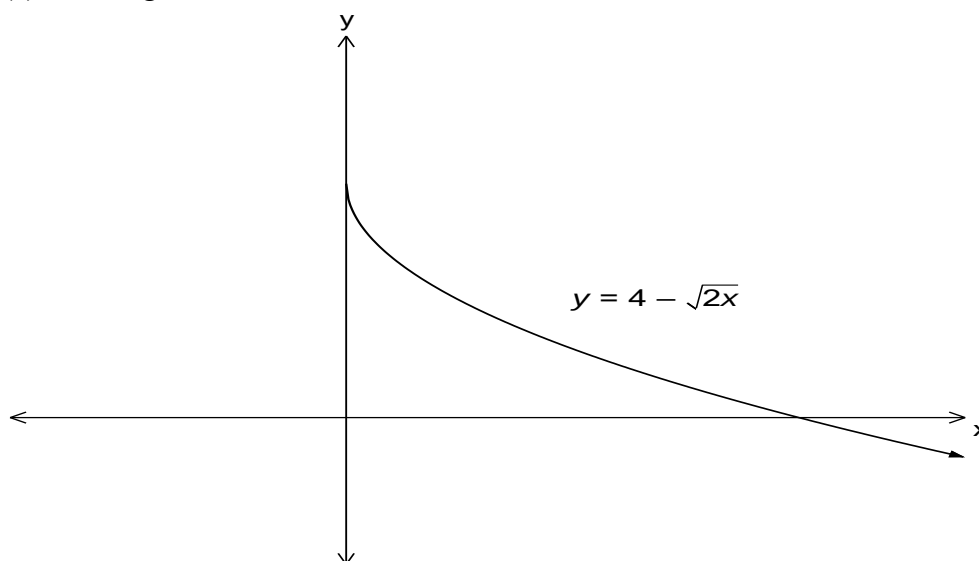
- (a) (i) State the period of the equation $y = 3 - \cos 2x$. (1)
 (ii) Graph $y = 3 - \cos 2x$ for $0 \leq x \leq \pi$. (3)

(b)



- (i) Show that the curves $y = \sin x$ and $y = \sqrt{3} \cos x$ intersect at $x = \frac{\pi}{3}$ and $x = \frac{4\pi}{3}$. (1)
 (ii) Hence determine the area enclosed between the curves for $0 \leq x \leq 2\pi$. (3)

- (c) The diagram below shows the function $y = 4 - \sqrt{2x}$.



- If the area enclosed by this curve $y = 4 - \sqrt{2x}$, the x -axis and the y -axis is rotated about the y -axis, find the volume of the solid of revolution formed. (4)

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Question 7 (Answer in a separate booklet)

- (a) The mass M kg of a radioactive substance present after t years is given by

$$M = 10e^{-kt}$$

where k is a positive constant. After 100 years the mass has been reduced to 5 kg.

- (i) What was the initial mass? (1)
- (ii) Find the value of k . (1)
- (iii) What amount of radioactive substance would remain after a period of 1000 years? (1)
- (iv) How long would it take for the initial mass to reduce to 8kg? (2)

- (b) A student attaches an air pressure pump to his bike tyre to inflate the tyres.
The rate of change of the pressure in the tyre is given by

$$\frac{dP}{dt} = \frac{t}{2} - \frac{1}{8} \text{ units of pressure/sec}$$

where t is the time after pumping has started.

After 4 seconds of pumping, the pressure in the tyre is 15 units.

- (i) What is the initial pressure in the tyre? (2)
- (ii) The bike tyre will burst when the pressure reaches 40 units of pressure.
At what time does this occur? (answer to the nearest second). (2)

- (c) Determine the values of x in the following equation. (3)

$$3 - 4\cos^2(2x) = 0 \quad 0 \leq x \leq \pi$$

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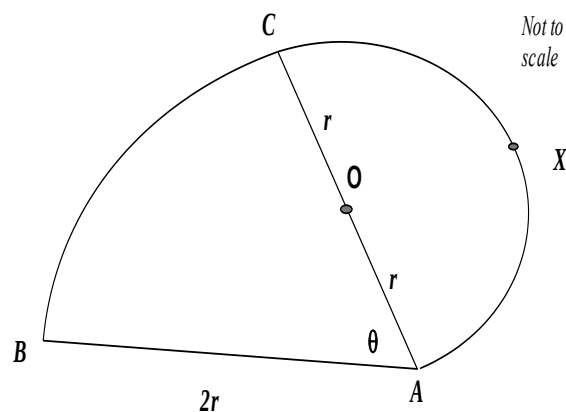
Question 8 (Answer in a separate booklet)

- (a) (i) A is the point $(1, 1)$ and B is the point $(4, 1)$. A point $P(x, y)$ moves such that $PA \perp PB$.

Show that the locus of P is given by $x^2 - 5x + y^2 - 2y = -5$. (3)

- (ii) Give a detailed geometric description of the locus in part (i). (2)

- (b) The cross-section of a metal object is formed as shown in the figure below.



The cross-section consists of a semi-circle AXC , centred at O and radius r , and a sector ABC of radius $2r$, centred at A with angle θ .

- (i) What is the perimeter $AXCB$ of the object in terms of r and θ ? (2)

- (ii) If the area of the cross-section is fixed at 4 square units, show that the perimeter in part (i) can also be expressed as (2)

$$P = \frac{4}{r} + r \left(2 + \frac{\pi}{2} \right)$$

- (iii) Show that the perimeter in part (ii) is smallest when $r^2 = \frac{8}{4 + \pi}$. (3)

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Question 9 (Answer in a separate booklet)

- (a) The volume formed by rotating the curve $y = \log_e x$ about the x – axis between $x = 2$ and $x = 6$ can be expressed as:

$$\text{Volume} = \pi \int_2^6 (\log_e x)^2 dx$$

Use Simpson's Rule with three function values to find an approximation for the volume of the solid of revolution formed. (3)

- (b) A particle is moving in a straight line. Its displacement, x metres, from the origin, O , at time t seconds, where $t \geq 0$, is given by

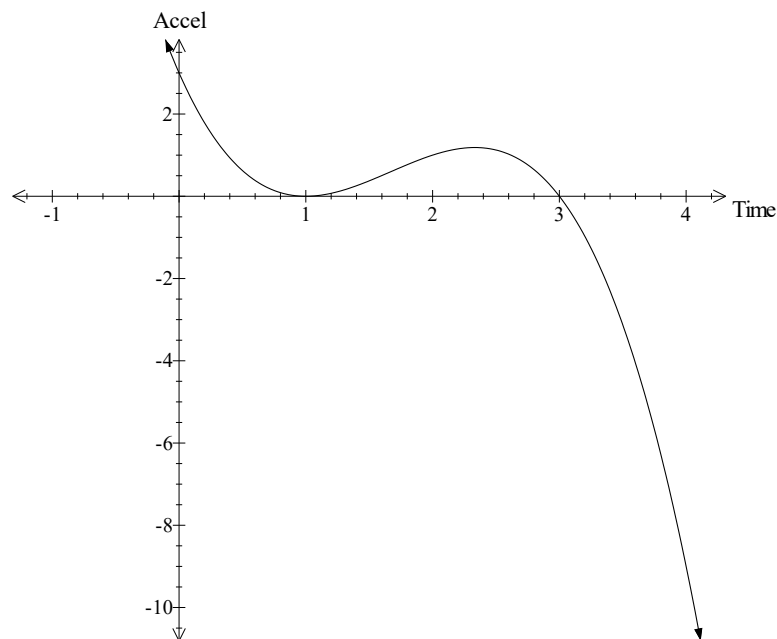
$$x = 4 + 3te^{-2t}$$

(i) Show the velocity of the particle is given by $\dot{x} = 3e^{-2t}(1 - 2t)$. (2)

(ii) Show the particle is at rest when $t = \frac{1}{2}$. (1)

(iii) Find the greatest possible total distance the particle could travel. (3)

- (c) An object is moving on the x – axis. The graph below shows the acceleration $a = \frac{d^2x}{dt^2}$.



Draw a velocity-time graph for $0 \leq t \leq 4$ if the object is initially travelling to the left with a speed of 3 m/s. (3)

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Question 10 (Answer in a separate booklet)

12

Consider the function $y = (x - 1) \log_e 2x$.

(i) What is the domain of this function? (1)

(ii) Find the x -intercepts of the function. (2)

(iii) Show that the gradient of the curve is given by $\frac{dy}{dx} = \frac{x-1}{x} + \log_e 2x$. (1)

(iv) Determine the gradient of the function at $x = \frac{1}{2}$ and at $x = 1$. (2)

(v) Without further calculation, determine whether the function has a maximum or minimum value. Justify your conclusion. (2)

(vi) By using any of the results above, show that $\int_{\frac{1}{2}}^1 \log_e 2x \, dx = \log_e 2 - \frac{1}{2}$ (4)

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

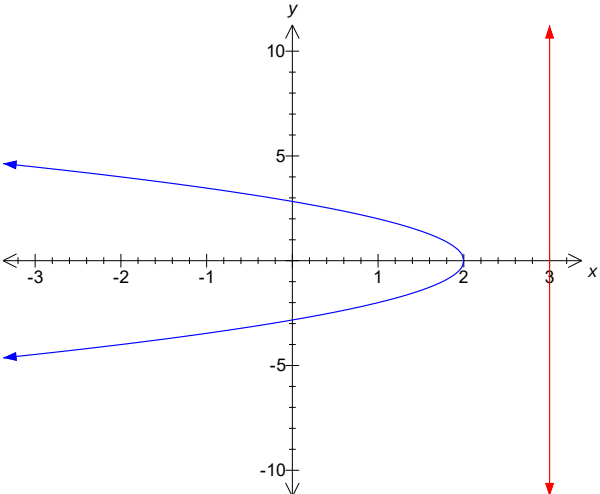
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Question 1

a)	$2(4x - 1) - 13x = 8 - 2 - 13x$ $= -5x - 2$	1 correct solution 0 marks if then solved as an equation
b)	7, 1, -5 $d = 1 - 7 \quad d = -5 - 1 \quad a = 7$ $= -6 \quad = -6$ $T_n = a + (n - 1)d$ $= 7 + (n - 1) \times -6$ $= 7 - 6n + 6$ $= 13 - 6n \text{ (as required)}$	2 correct solution 1 either showing $T_n = 7 + (n - 1) \times -6$ OR $d = -6$
c)	$ 2x - 5 > 3$ $2x - 5 > 3 \quad \text{or} \quad -(2x - 5) > 3$ $2x > 8 \quad -2x + 5 > 3$ $x > 4 \quad -2x > -2$ $x < 1$ 	3 solution correct 2 both solutions but no number line 1 only one correct solution
d)	$y = \cos^3 x$ $= (\cos x)^3$ $\frac{dy}{dx} = 3(\cos x)^2 \times -\sin x$ $= -3\sin x \cos^2 x$	2 correct solution 1 any one of $-, 3, \sin x$ or $\cos^2 x$ absent from correct solution
e)		1 correct graph showing both x and y intercepts
f)	$\frac{5}{2 + \sqrt{3}} = a + b\sqrt{3}$ $\text{LHS} = \frac{5}{2 + \sqrt{3}} \times \frac{2 - \sqrt{3}}{2 - \sqrt{3}}$ $= \frac{10 - 5\sqrt{3}}{4 - 3}$ $= 10 - 5\sqrt{3} \quad \therefore a = 10, b = -5$	2 both a and b correct 1 multiplying numerator and denominator by $2 - \sqrt{3}$
g)	$\log_5 125 = \log_5 5^3$ $= 3\log_5 5$ $= 3 \times 1$ $= 3$	1 correct solution

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Question 2

(a)	For real roots, $\Delta \geq 0$ $(-4)^2 - 4 \times 2k \geq 0$ $16 - 8k \geq 0$ $-8k \geq -16$ $k \leq 2 \text{ as required}$	2 marks correct solution 1 mark for getting to line 3 Note: taking a few particular values of k and substituting in is not a general proof
(b) (i)	$\int x^{\frac{1}{3}} + x^2 dx = \frac{3}{4}x^{\frac{4}{3}} + \frac{x^3}{3} + c \text{ or } \frac{3}{4}x\sqrt[3]{x} + \frac{x^3}{3} + c$	2 marks- correct integral including +c 1 mark – 1 error in answer Note: Many students need to revise index laws so the can rewrite $\sqrt[3]{x}$ correctly in index form
(ii)	$\frac{1}{2}e^{2x} - \frac{1}{3}\cos 3x + c$	2 marks – correct answer 1 mark – one integral correctly found Note: Students should use the Table of Standard Integrals to correctly find coefficients
(c)	Rewrite given equation as $y^2 = -4(x-2)$ is of the form $(y-k)^2 = -4a(x-h) \text{ where } a = 1$ Vertex is $(h,k) = (2,0)$  Directrix is $x=3$	2 marks for correct vertex and directrix equation 1 mark for correct vertex OR correct directrix from incorrect vertex NOTE!!!!: the focal length is a LENGTH so stating that $a=-1$ (as many did!!) is VERY WRONG. Main other error was in not rearranging terms to correct form which requires x to be first term in bracket.
(d)	$\frac{dy}{dx} = 3\sin 2x$ $y = -\frac{3}{2}\cos 2x + c$ $(0,0)$ $0 = -\frac{3}{2}\cos 0 + c \therefore c = \frac{3}{2}$ $\therefore y = -\frac{3}{2}\cos 2x + \frac{3}{2}$	2 marks- correct solution 1 mark correct Line 2 OR incorrect integral but correct finding of constant from error.
(e)	$\tan \theta = -2$ Related acute $= 63^\circ 26' = 1.107148718^\circ$ $\theta = \pi - 1.107148718^\circ \text{ or } \theta = 2\pi - 1.107148718^\circ$ $\theta \approx 2.03 \text{ or } 5.18$	2 marks – correct solutions, including reasonable approximations in terms of π 1 mark – correct answer in degrees OR 1 correct answer in radians OR correct radians but in first or third quadrants OR correct radian answers from making -2 the radians

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Question 3

(ai)	(0,2)	1 mark – correct answer must STATE CO-ORDINATES
(aii)	$y = \frac{3}{4}x - 3$	1 mark – correct answer
(aiii)	(4,0)	1 mark – correct answer
(aiv)	CD must be same length as AB and parallel to it, so D is (4,5)	1 mark – correct answer
(av)	$L_2: 3x - 4y - 12 = 0 \quad A(0, 2)$ $d_1 = \frac{ ax_1 + by_1 + c }{\sqrt{a^2 + b^2}}$ $d_1 = \frac{ 3 \times 0 - 4 \times 2 - 12 }{\sqrt{3^2 + 4^2}}$ $d_1 = \frac{ -20 }{5}$ $d_1 = 4$	1 mark – correct answer
(avi)	A rhombus is also a parallelogram $A = bh_{\perp}$ $= 5 \times 4$ $= 20 \text{ square units}$	2 marks – correct area or correct from error in (v) 1 mark – correct length of BC or AD
(b)	$100 = \alpha + 30$ (exterior angle of triangle AEB equals sum of interior opposite angles) $\alpha = 70$ $40 + 70 + \beta = 180$ (angle sum of triangle ACD) $\beta = 70$	2 marks – correct angles with correct reasons given for the approach taken 1 mark – correct angle with correctly argued reasons
(c)	In $\triangle BAC$ and $\triangle DAE$, $\frac{AB}{AE} = \frac{6}{18} = \frac{1}{3}$ $\frac{AC}{AD} = \frac{4}{12} = \frac{1}{3} \quad \therefore \frac{AB}{AE} = \frac{AC}{AD}$ $\angle BAC$ is a common angle $\therefore \triangle BAC \parallel \triangle DAE$ (two pairs of corresponding sides in proportion about the same INCLUDED angle)	3 marks – correctly argued proof 2 marks – either omitting test used to prove similarity OR incorrect/inadequate wording of test OR not directly identifying names of corresponding sides (only referring to lengths) OR only giving test reason and omitting which triangle are similar 1 mark – finding common angle OR giving numerical values of correct corresponding sides

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Question 4

(a)	$2x^2 + 6x + 9 \equiv A(x+2)(x-1) + B(x+2) + C$ coefficient of x^2 $A = 2$ let $x = -2$ $2 \times (-2)^2 + 6 \times (-2) + 9 = 0 + 0 + C$ $C = 5$ let $x = 1$ $2 \times 1^2 + 6 \times 1 + 9 = 0 + 3B + 5$ $B = 4$	3 marks – one mark for each correct answer.
(b) (i)	$T_1 = 3 \quad T_2 = 6 \quad T_3 = 12$ $\frac{T_2}{T_1} = \frac{T_3}{T_2} = 2 = r$ $T_8 = ar^{n-1}$ $= 3 \times 2^7$	1 mark – correct answer
(b) (ii)	$T_{15} = 3 \times 2^{14}$ $S_{15} = \frac{a(r^n - 1)}{r - 1}$ $= \frac{3(2^{15} - 1)}{2 - 1}$ $= 98301$	1 mark – correct answer – non-index form accepted.
(c)	$T_1 = 133$ $T_n = 133 + (n - 1) \times 7$ $= 126 + 7n$ $126 + 7n < 500$ $7n < 374$ $n < 53.43$ $n = 53$ $T_{last} = 126 + 7 \times 53$ $= 497$ $S_n = \frac{n}{2}(a + l)$ $= \frac{53}{2}(133 + 497)$ $= 16695$	4 marks – correct solution 3 marks – correct value for last term in series 2 marks – correct value for n 1 mark – expression for T_n

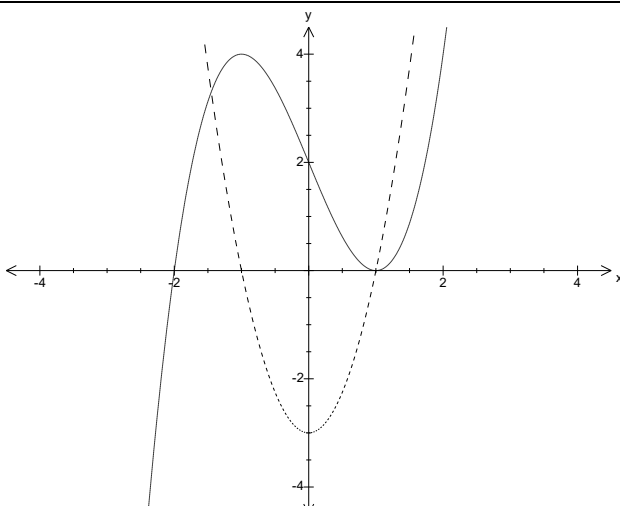
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Question 4 (continued)

(d) $S_n = 1 + \frac{\sin\theta}{2} + \frac{\sin^2\theta}{4} + \frac{\sin^3\theta}{8} + \dots$ <i>Geometric Series</i> $r = \frac{\sin\theta}{2}$ $-1 \leq \sin\theta \leq 1$ $\therefore -\frac{1}{2} \leq \frac{\sin\theta}{2} \leq \frac{1}{2} \therefore \text{Series will have limiting sum.}$ $S_n = \frac{a}{1-r}$ $= \frac{1}{1 - \frac{\sin\theta}{2}}$ S_n is maximum when denominator is min. ie. $\frac{\sin\theta}{2}$ is at its max value $S_n = \frac{1}{1 - \frac{1}{2}}$ $= 2$	3 marks – correct answer. 2 marks – correct expression for S_n 1 mark – identify series as geometric series and determining value of r will mean that it has a limiting sum.
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Question 5

(a)(i)	$a = 2 \quad b = -3 \quad c = 6$ $\alpha + \beta = -\frac{b}{a} = -\frac{-3}{2} = \frac{3}{2}$	1 mark – correct answer								
(ii)	$\alpha \beta = \frac{c}{a} = 3$	1 mark – correct answer								
(iii)	$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha \beta$ $= \frac{9}{4} - 6 = -\frac{15}{4}$	2 marks – correct answer 1 mark correct expression								
(b)	Cannot use calculus on trig functions expressed in degrees – must be radians. $y = \sin 2x^\circ = \sin \frac{2\pi}{180}x$ $\therefore \frac{dy}{dx} = \frac{2\pi}{180} \cos \frac{2\pi}{180}x = \frac{\pi}{90} \cos \frac{\pi x}{90}$	2 marks – correct derivative 1 mark – recognition of need for radians NOTE: When using calculus with trig functions, you must use RADIANS.								
(c)		2 marks – correct graph 1 mark – correct x-intercepts In many diagrams, the two curves were hard to distinguish. Use different colours, dotted lines, etc to identify each and label.								
(d)	$y = 12x^3 - 3x^4$ $\frac{dy}{dx} = 36x^2 - 12x^3$ For SP $\frac{dy}{dx} = 0$ $\therefore 12x^2(3 - x) = 0$ $x = 0 \text{ or } x = 3$ $\frac{d^2y}{dx^2} = 72x - 36x^2$ At $x = 0 \frac{d^2y}{dx^2} = 0$ so possible POI At $x = 3 \frac{d^2y}{dx^2} = -108 < 0 \therefore \text{max}$ $\therefore$ SP at (0, 0) POI (3, 81) maximum value <table border="1" data-bbox="181 1957 549 2024"><tr><td>x</td><td>-1</td><td>0</td><td>1</td></tr><tr><td>y''</td><td>-108</td><td>0</td><td>36</td></tr></table>	x	-1	0	1	y''	-108	0	36	4 marks – correct points and nature determined 3 marks – correct points but incomplete analysis of second derivative or no y values. 2 marks – correct first derivative and x values determined 1 mark – correct derivatives
x	-1	0	1							
y''	-108	0	36							

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Question 6

a) i)	$\text{i) period} = \frac{2\pi}{2}$ $= \pi$	1 correct answer
ii)	$y = 3 - \cos 2x$	3 correct graph 2 $y = -\cos 2x$ at start other than 2 on y-axis or $y = 3 + \cos 2x$ or $y = 2 - \cos x$ 1 $y = \cos 2x$
b) i)	$x = \frac{\pi}{3} \quad y = \sin\left(\frac{\pi}{3}\right) \quad y = \sqrt{3} \cos\left(\frac{\pi}{3}\right)$ $= \frac{\sqrt{3}}{2} \quad = \frac{\sqrt{3}}{2}$ $x = \frac{4\pi}{3} \quad y = \sin\left(\frac{4\pi}{3}\right) \quad y = \sqrt{3} \cos\left(\frac{4\pi}{3}\right)$ $= -\sin\left(\frac{\pi}{3}\right) \quad = -\sqrt{3} \cos\left(\frac{\pi}{3}\right)$ $= -\frac{\sqrt{3}}{2} \quad = -\frac{\sqrt{3}}{2}$ OR $\sqrt{3} \cos x = \sin x$ $1 = \sin \frac{x}{\sqrt{3} \cos x}$ $1 = \frac{1}{\sqrt{3}} \tan x$ $\tan x = \sqrt{3} \quad \text{1st quadrant } \angle \frac{\pi}{3}$ $\tan x > 0 \text{ quad 1 and 3}$ $x = \frac{\pi}{3}, \frac{4\pi}{3}$	1 showing intersection at both points OR using tan to find the 2 solutions

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Question 6 (continued)

ii)	$Area = \int_{\frac{\pi}{3}}^{\frac{4\pi}{3}} (\sin x - \sqrt{3} \cos x) dx$ $= [-\cos x - \sqrt{3} \sin x]$ $= \left(-\cos \left(\frac{4\pi}{3} \right) - \sqrt{3} \sin \left(\frac{4\pi}{3} \right) \right) - \left(-\cos \left(\frac{\pi}{3} \right) - \sqrt{3} \sin \left(\frac{\pi}{3} \right) \right)$ $= \left(-\frac{1}{2} - \sqrt{3} \times -\frac{\sqrt{3}}{2} \right) - \left(-\frac{1}{2} - \sqrt{3} \times \frac{\sqrt{3}}{2} \right)$ $= \left(\frac{1}{2} + \frac{3}{2} \right) - \left(-\frac{1}{2} - \frac{3}{2} \right)$ $= \left(\frac{4}{2} \right) - \left(-\frac{4}{2} \right)$ $= 2 + 2$ Area is 4 units²	3 correct solution 2 1 of the integral sections correct with correct solution for the incorrect integral 11 correct integral only OR 2 integrals incorrect with incorrect limits OR area broken into 3 regions with correct answer of 4 units² for the only correct region
c)	y intercept when $x = 0$ $y = 4$ $y = 4 - \sqrt{2x}$ $\sqrt{2x} = 4 - y$ $2x = (4 - y)^2$ $x = \frac{(4 - y)^2}{2}$ $x^2 = \frac{(4 - y)^4}{4}$ $Volume = \pi \int_0^4 \frac{(4 - y)^4}{4} dy$ $= \frac{\pi}{4} \left[\frac{(4 - y)^5}{5 \times -1} \right]_4^0 \quad ***$ $= -\frac{\pi}{20} \{ (4 - 4)^5 - (4 - 0)^5 \}$ $= -\frac{\pi}{20} \times -1024$ $= \frac{256\pi}{5} \text{ units}^3$	4 correct solution 3 correct integral to *** OR no pi in solution. 2 correct value to x² 1 correct y intercept

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Question 7

(a)-(i)	$M = 10e^{-kt}$ At $t = 0$ $M = 10e^{-k \times 0}$ $M = 10$ therefore initial mass is 10kg	1 mark – correct answer
(ii)	$M = 10e^{-kt}$ at $t = 100$ $m = 5$ $5 = 10e^{-100k}$ $\frac{\ln\left(\frac{1}{2}\right)}{-100} = k$ $k = \frac{\ln 2}{100} = 6.93 \times 10^{-3}$	1 mark – correct answer
(iii)	$M = 10e^{-1000k}$ $= 9.765 \times 10^{-3} \text{ kg}$	1 mark – correct answer
(iv)	$8 = 10e^{-kt}$ $t = \frac{\ln(0.8)}{-k}$ $t = 32.19 \text{ year}$	2 mark – correct answer 1 mark – correct approach with arithmetic error
(b)-(i)	$\frac{dP}{dt} = \frac{t}{2} - \frac{1}{8}$ $P = \frac{t^2}{4} - \frac{t}{8} + C$ $t = 4$ $P = 15$ $15 = \frac{4^2}{4} - \frac{4}{8} + C$ $C = 11.5$ $P = \frac{t^2}{4} - \frac{t}{8} + 11.5$ At $t = 0$ $P = 11.5$ Therefore initial pressure is 11.5 units.	2 marks – correct answer 1 mark – correct integration with C

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Question 7 (continued)

(b)-(ii)	$P = \frac{t^2}{4} - \frac{t}{8} + 11.5$ $40 = \frac{t^2}{4} - \frac{t}{8} + 11.5$ $320 = 2t^2 - t + 92$ $t = \frac{1 \pm \sqrt{1825}}{4} = 10.9 \text{ sec} \approx 11 \text{ sec} \quad \text{as } t \geq 0$	2 marks – correct answer 1 mark – correct approach reaching two possible answers (one neg and one positive) with one arithmetic error.
(c)	$3 - 4\cos^2(2x) = 0 \quad 0 \leq x \leq \pi$ $\cos(2x) = \pm \frac{\sqrt{3}}{2} \quad 0 \leq 2x \leq 2\pi$ $2x = \frac{\pi}{6}, \frac{2\pi}{6}, \frac{4\pi}{6}, \frac{5\pi}{6}$ $x = \frac{\pi}{12}, \frac{\pi}{12}, \frac{2\pi}{12}, \frac{5\pi}{12}$	3 marks – correct answer 2 marks – incomplete domain 1 mark – failure to recognise $\pm$

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Question 8

(a) (i)	$m_{PA} \times m_{PB} = -1$ $\frac{y-1}{x-1} \times \frac{y-1}{x-4} = -1$ $(y-1)^2 = -(x-1)(x-4)$ $y^2 - 2y + 1 + (x-1)(x-4) = 0$ $y^2 - 2y + 1 + x^2 - 5x + 4 = 0$ $x^2 - 5x + y^2 - 2y = -5$	3 marks – correct demonstration 2 marks – product of gradients expressed correctly but subsequent error 1 mark – both gradients correct
(ii)	Completing the square: $\left(x - \frac{5}{2}\right)^2 - \frac{25}{4} + (y-1)^2 - 1 + 5 = 0$ $\left(x - \frac{5}{2}\right)^2 + (y-1)^2 = \left(\frac{3}{2}\right)^2$ The locus is a circle centred at $\left(\frac{5}{2}, 1\right)$ with radius $\frac{3}{2}$	2 marks – complete description 1 mark – squares completed correctly.
(b) (i)	Perimeter of semi-circle $AXC = \pi r$ CB Arc length $= 2r\theta$ Hence perimeter $AXCB = \pi r + 2r\theta + 2r$	2 marks – correct perimeter 1 mark – at least one of the first two quantities correct
(bii)	$Area = A_{\text{sector}} + A_{\text{semi circle}}$ $4 = \frac{1}{2}(2r)^2\theta + \frac{1}{2}\pi r^2$ $4 = 2r^2\theta + \frac{1}{2}\pi r^2$ $2r^2\theta = 4 - \frac{1}{2}\pi r^2$ $\theta = \frac{2}{r^2} - \frac{\pi}{4}$ $\therefore P = 2r + \pi r + 2r\left(\frac{2}{r^2} - \frac{\pi}{4}\right)$ $P = 2r + \pi r + \frac{4}{r} - \frac{\pi r}{2}$ $P = \frac{4}{r} + 2r + \frac{\pi r}{2}$ $P = \frac{4}{r} + r\left(2 + \frac{\pi}{2}\right)$	2 marks – correct proof 1 mark – correct expression for θ substituted into P

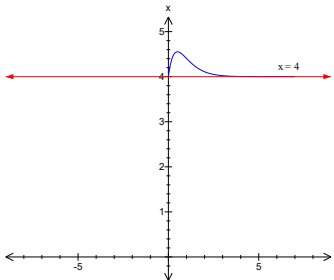
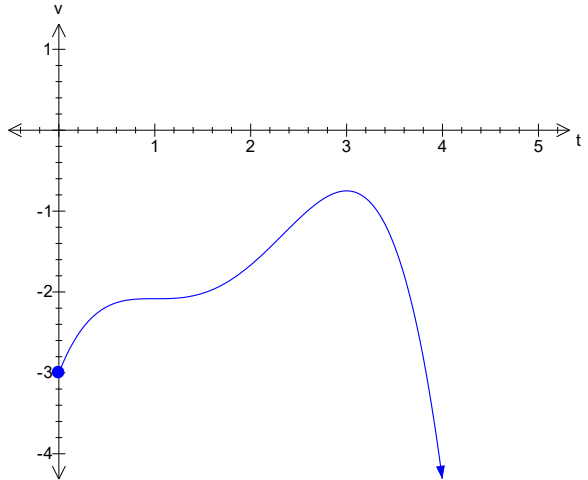
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Question 8 (continued)

(iii)	$P = 4r^{-1} + 2r + \frac{\pi r}{2}$ $\frac{dP}{dr} = -4r^{-2} + 2 + \frac{\pi}{2}$ $\frac{d^2P}{dr^2} = 8r^{-3} > 0 \text{ for all } r > 0 \text{ so minimum}$ For min $\frac{dP}{dr} = 0$ $\therefore \frac{4}{r^2} = 2 + \frac{\pi}{2}$ $r^2 = \frac{8}{4 + \pi}$	<i>3 marks – correct and complete demonstration</i> <i>2 marks – correct derivative and expression for r^2 but subsequent error or no proof of minimum</i> <i>1 mark – correct derivative, no proof of a minimum or algebraic error.</i>
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Question 9

(a)	$\text{Vol} = \pi \int_2^6 (\log_e x)^2 dx \approx \frac{(6-2)\pi}{6} [(\ln 2)^2 + 4(\ln 4)^2 + (\ln 6)^2]$ $= \frac{2\pi}{3} \times 11.37810323$ $= 23.83 u^3$	3 marks – correct solution 2 marks – correct Simpson's rule except for loss of π or incorrect or omitted y^2 or calculation error 1 mark – a correct use of Simpson's rule on $\int_2^6 \log_e x dx$ NOTE: many students misinterpreted $(\log_e x)^2$
(b) (i)	$\dot{x} = 3t - 2e^{-2t} + e^{-2t} \cdot 3$ $= 3e^{-2t}(1 - 2t)$	2 marks for correct velocity expression 1 mark for correct application of product rule
(b) (ii)	Either substitute $t = \frac{1}{2}$, $3e^{-2 \times \frac{1}{2}} \left(1 - 2 \times \frac{1}{2}\right) = 0$ i.e. particle is at rest. $e^{-2t}(1 - 2t) = 0 \quad \therefore e^{-2t} = 0 \text{ or } 1 - 2t = 0$ OR Solve $e^{-2t} \neq 0$ as $e^{-2t} > 0$ $t = \frac{1}{2}$ as req'd	1 mark – either correct solution NOTE: many students either did not write a conclusion for the substitution approach OR ignored the possibility of $e^{-2t} = 0$ OR omitted a justification for not solving
(b) (iii)	When $t=0$, $x=4$ When $t=0.5$, $x = 4 + \frac{3}{2e}$ As $t \rightarrow \infty$, $e^{-2t} \rightarrow 0 \therefore x \rightarrow 4$  Max distance = $2 \times \frac{3}{2e} = \frac{3}{e}$ metres	3 marks – correct solution 2 marks – establishing starting position, maximum and limit 1 mark – finding max displacement NOTE: many students believed the max displacement represented the greatest possible total distance
(c)	Graph had to start at (0,-3), have a point of inflection when $t=1$, and a rel max at $t=3$ 	3 marks – correct graph 2 marks – 1 error 1 mark - 2 errors

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Question 10

(a) (i)	$y = (x-1) \log_e 2x$ Domain of $(x-1)$ is all real x while the domain of $\ln x$ is $x > 0$. Hence the domain of the product is $x > 0$.	1 mark – correct domain
(ii)	x -intercepts when $y = 0$. Hence $x = 1$ OR $\ln 2x = 0$ So $x = 0.5$	2 marks – both intercepts correct 1 mark – one intercept correct
(iii)	$\frac{dy}{dx} = \ln 2x \times 1 + (x-1) \times \frac{1}{x}$ $= \frac{x-1}{x} + \ln 2x$	1 mark – correct derivative
(iv)	At $x = \frac{1}{2}$, $\frac{dy}{dx} = \frac{-\frac{1}{2}}{\frac{1}{2}} + \ln 2 \times \frac{1}{2} = -1$ At $x = 1$ $\frac{dy}{dx} = \ln 2$	2 marks – both gradients correct 1 mark – one gradient correct.
(v)	The curve has a minimum as the gradient changes from negative to positive and is continuous throughout the sub-domain stated.	2 marks - statement of a minimum supported with both statements 1 mark statement of a minimum with some appropriate support
(vi)	From part (iii) $\frac{d}{dx} \{(x-1)\ln 2x\} = 1 - \frac{1}{x} + \ln 2x$ $\therefore \int_{\frac{1}{2}}^1 \frac{d}{dx} \{(x-1)\ln 2x\} = \int_{\frac{1}{2}}^1 1 dx - \int_{\frac{1}{2}}^1 \frac{1}{x} dx + \int_{\frac{1}{2}}^1 \ln 2x dx$ $\left[(x-1)\ln 2x \right]_{\frac{1}{2}}^1 = \left[x - \ln x \right]_{\frac{1}{2}}^1 + \int_{\frac{1}{2}}^1 \ln 2x dx$ $\therefore \int_{\frac{1}{2}}^1 \ln 2x dx = (0-0) - \left[(1-0) - \left(\frac{1}{2} - \ln \frac{1}{2} \right) \right]$ $= -\frac{1}{2} - \ln 1 + \ln 2 = \ln 2 - \frac{1}{2}$	4 marks – correct process to demonstrate required result 3 marks – correct up to final substitution 2 marks – integral and limits mostly correct 1 mark – correct setting out for integral using the derivative